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TREND-BASED MOCK PAPER

GATE Textile Engineering and Fibre Science

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Textile Engineering and Fibre Science
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Perimeter of square side 5:

(Numerical Answer Type — Enter the numerical value)

Section B: Textile Engineering and Fibre Science

Q2. [1-MCQ] [1 mark]

Integral of $2x \, dx$:

- (A) $x^2 + C$
 - (B) $2x + C$
 - (C) $x + C$
 - (D) $x^{2/2}$
-

Q3. [1-NAT] [1 mark]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
 - (B) 0.092
 - (C) 0.762
 - (D) 0.195
-

Q5. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$

- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q6. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q7. [1-MCQ] [1 mark]

The vector (1,2,3) projected onto (2,1,0) is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) (2,1,0)
- (D) (1,2,3)

Q8. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q9. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q10. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q11. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q12. [1-MCQ] [1 mark]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q13. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q14. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q15. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q16. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q17. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q19. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q20. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3

(D) 0

Q21. [2-MSQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q22. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q23. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q24. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q25. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q26. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MSQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q28. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{(\|u\| * \|v\|)}$

Q29. [2-NAT] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi * r^2$
- (B) $2 * \pi * r$
- (C) $\pi * r$
- (D) $2 * \pi * r^2$

Q33. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q34. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{(\|u\| * \|v\|)}$

Q36. [2-MCQ] [2 marks]

The Laplace transform of $t * e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q37. [1-MCQ] [1 mark]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q38. [1-MCQ] [1 mark]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q39. [2-NAT] [2 marks]

Weaving interlaces:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q41. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q42. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q43. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q44. [2-NAT] [2 marks]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q45. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q46. [2-NAT] [2 marks]

Weaving interlaces:

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q48. [2-NAT] [2 marks]

Weaving interlaces:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

Q50. [1-MCQ] [1 mark]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q51. [1-MCQ] [1 mark]

Wool is classified as:

- (A) Vegetable
- (B) Animal
- (C) Mineral
- (D) Synthetic

Q52. [1-MCQ] [1 mark]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q53. [1-MCQ] [1 mark]

Nylon is a:

- (A) Natural
- (B) Synthetic polymer
- (C) Regenerated
- (D) Mineral

Q54. [2-MCQ] [2 marks]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q55. [2-MCQ] [2 marks]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q56. [2-NAT] [2 marks]

Nylon is a:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

Wool is classified as:

- (A) Vegetable
- (B) Animal
- (C) Mineral
- (D) Synthetic

Q58. [2-MCQ] [2 marks]

Wool is classified as:

- (A) Vegetable
- (B) Animal
- (C) Mineral
- (D) Synthetic

Q59. [1-NAT] [1 mark]

Carding does:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q61. [1-NAT] [1 mark]

Tensile strength measured by:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

Q63. [1-MCQ] [1 mark]

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

Q64. [1-NAT] [1 mark]

Ring frame produces:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Coulomb friction law: $F = ?$

- (A) μN
 - (B) μ
 - (C) N/μ
 - (D) $\mu' N$
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1NAT	A	1
5	1MCQ	A	1
6	1NAT	B	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1NAT	C	1
10	1MCQ	A	1
11	1MCQ	A	1
12	1MCQ	B	1
13	1MCQ	B	1

14	1NAT	A	1
15	1MCQ	A	1
16	1MCQ	A	1
17	1MCQ	A	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1MCQ	B	1
21	1MCQ	B	1
22	1MCQ	C	1
23	1MCQ	A	1
24	1MCQ	C	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1MCQ	C	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1MCQ	C	1
31	2NAT	B	2
32	2MCQ	A	2
33	2MCQ	A	2
34	2MCQ	B	2
35	2MSQ	B	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2NAT	A	2
41	2MSQ	A	2
42	2MCQ	A	2
43	2NAT	A	2

44	2MCQ	A	2
45	2MCQ	A	2
46	2NAT	C	2
47	2NAT	C	2
48	2MCQ	A	2
49	2MCQ	B	2
50	2MCQ	A	2
51	2NAT	B	2
52	2MCQ	A	2
53	2NAT	A	2
54	2NAT	A	2
55	2MCQ	A	2
56	2MCQ	B	2
57	2MCQ	A	2
58	2NAT	A	2

59	2MCQ	B	2
60	2MCQ	B	2
61	2NAT	B	2
62	2MCQ	A	2
63	2MCQ	A	2
64	2NAT	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Fibers | Topic: Natural Fibers | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Yarn Manufacturing | Topic: Spinning | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Fabric Manufacturing | Topic: Weaving | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Testing | Topic: Properties | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Chemical Processing | Topic: Dyeing | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Apparel Engineering | Topic: Pattern Making | Difficulty: moderate

Q9.

Answer: C

Solution:

Correct Answer: C

Subject: Textile Machinery | Topic: Machines | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Textile Physics | Topic: Friction | Difficulty: difficult

Q11.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q12.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q13.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q14.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q15.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q16.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q18.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x \cdot v$ and $y(1) = 0$ gives $v = 0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1 + 4 + 9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q20.

Answer: B

Solution:

$\text{proj} = ((v \cdot w) / (w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q22.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q25.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q26.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q31.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q40.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q44.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q45.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q46.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q47.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q49.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q50.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q51.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q52.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q53.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q54.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q55.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \text{ Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q56.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q57.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q58.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q59.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q60.

Answer: B

Solution:

$$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: B

Solution:

$f(0)=2, f(1)=0, f(-1)=4, f(2)=4, f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q62.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q63.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q64.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q65.

Answer: A

Solution:

$$\mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds}(F(s)). \quad F(s) = \frac{1}{(s+2)}, \quad -\frac{d}{ds}\left[\frac{1}{(s+2)}\right] = \frac{1}{(s+2)^2}.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 35 questions, 55 marks
Textile Manufacture: 12 questions, 22 marks
Textile Fibers: 10 questions, 15 marks
General Aptitude: 1 questions, 1 marks
Yarn Manufacturing: 1 questions, 1 marks
Fabric Manufacturing: 1 questions, 1 marks
Textile Testing: 1 questions, 1 marks
Chemical Processing: 1 questions, 1 marks
Apparel Engineering: 1 questions, 1 marks
Textile Machinery: 1 questions, 1 marks
Textile Physics: 1 questions, 1 marks

Difficulty Distribution:

Easy: 25 | Moderate: 33 | Difficult: 7

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.